Homogeneous Linear ODE's with constant coefficients

1. The General Form

For an n-th order differential equation, the homogeneous form looks like this:

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

- **Linear:** The dependent variable y and its derivatives are all to the power of 1 and are not multiplied together.

For second-order ODE,

A differential equation of the form $\boldsymbol{y'' + p(x)y' + q(x)y = r(x)}$ is called linear ODE.

This differential equation reduces to homogeneous if $\boldsymbol{r(x) = 0}$ and nonhomogeneous if $\boldsymbol{r(x) \neq 0}$

Examples:

Second order Linear ODE: $y'' + 25y = e^{-x}cosx$

$xy'' + y' + xy = x^2$

Second order Linear homogeneous ODE: $y'' + 25y = 0$

$xy'' + y' + xy = 0$

Second order Linear nonhomogeneous ODE: $y'' + 25y = e^{-x}cosx$

$xy'' + y' + xy = x^2$

General Form:

If $e^{m_1x}, e^{m_2x}, e^{m_3x}, \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, e^{m_nx}$ be the n independent solutions of the differential equation $P_0\frac{d^n y}{dx^n} + P_1\frac{d^{n-1}y}{dx^{n-1}} + P_2\frac{d^{n-2}y}{dx^{n-2}} + \ldots\ldots\ldots\ldots\ldots\ldots\ldots P_n y = 0$ then its general solution is $y = c_1e^{m_1x} + c_2e^{m_2x} + c_3e^{m_3x} + \ldots\ldots\ldots\ldots\ldots\ldots c_ne^{m_nx}$, where c_1, c_2, c_3 are arbitrary constants.

NB: The trial solution is assumed exponential trial solution because exponential functions keep the same form after differentiation.

✅ **Summary**

It is the general solution because:

1. Each $e^{m_i x}$ comes from a root of the characteristic equation.
2. Linear DEs allow superposition (sum of solutions is also a solution).
3. The solution contains n arbitrary constants, matching the order of the DE.

For second order ODE

If $e^{m_1 x}$ and $e^{m_2 x}$ be the two independent solutions of the differential equation $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$ then its general solution is $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$, where c_1 and c_2 are arbitrary constants.

Auxiliary equation:

Let the differential equation $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$ .....(i)

Let $y = e^{mx}$ be a solution of equation (i)

Then $\frac{dy}{dx} = me^{mx}$, $\frac{d^2y}{dx^2} = m^2 e^{mx}$

Then equation (i) becomes

$m^2 e^{mx} + Pme^{mx} + Qe^{mx} = 0$

or, $(m^2 + Pm + Q)e^{mx} = 0$

Since, $e^{mx} \neq 0$

Auxiliary equation is $m^2 + Pm + Q = 0$

Then three cases may arise:

Case I: The roots are real and unequal.

If $m = \alpha, \beta$ then the general solution is

$$y = c_1 e^{\alpha x} + c_2 e^{\beta x}$$

Case II: The roots are real and equal.

If $m = \alpha, \alpha$ then the general solution is

$$y = (c_1 + c_2 x) e^{\alpha x}$$

Case III: The roots are imaginary.

If $m = \alpha \pm i\beta$ then the general solution is

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

Ex-1: Solve $\frac{d^2 y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$

Solu: Given equation is $\frac{d^2 y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ .....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore \frac{dy}{dx} = me^{mx}$, $\frac{d^2 y}{dx^2} = m^2 e^{mx}$

Then equation (i) becomes

$$m^2 e^{mx} + 5me^{mx} + 6e^{mx} = 0$$

or, $(m^2 + 5m + 6)e^{mx} = 0$

Since, $e^{mx} \neq 0$

The auxiliary equation is

$$m^2 + 5m + 6 = 0$$

$\therefore\ m = -3, -2$

General solution is $y = c_1 e^{-3x} + c_2 e^{-2x}$

Ex-2: Solve $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 4y = 0$

Solu: Given equation is

$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 4y = 0$ .....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2 e^{mx}$

Then equation (i) becomes

$m^2 e^{mx} + 6me^{mx} + 4e^{mx} = 0$

or, $(m^2 + 6m + 4)e^{mx} = 0$

Since, $e^{mx} \neq 0$,

The auxiliary equation is

$m^2 + 6m + 4 = 0$

$\therefore\ m = \frac{-6 \pm \sqrt{6^2 - 4.1.4}}{2.1}$

$or, \qquad m = -3 \pm \sqrt{5}$

$\therefore$ General solution is

$y = c_1 e^{(-3+\sqrt{5})x} + c_2 e^{(-3-\sqrt{5})x}$

$or, \qquad y = e^{-3x}(c_1 e^{\sqrt{5}x} + c_2 e^{-\sqrt{5}x})$

Ex-3: Solve $2\frac{d^3y}{dx^3} - 3\frac{dy}{dx} + y = 0$

Solu: Given equation is

$2\frac{d^3y}{dx^3} - 3\frac{dy}{dx} + y = 0$ .....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2 e^{mx}, \quad \frac{d^3y}{dx^3} = m^3 e^{mx}$

Then equation (i) becomes

$2m^3 e^{mx} - 3me^{mx} + e^{mx} = 0$

or, $(2m^3 - 3m + 1)e^{mx} = 0$

Since, $e^{mx} \neq 0$

The auxiliary equation is

$2m^3 - 3m + 1 = 0$

$2m^3 - 2m^2 + 2m^2 - 2m - m + 1 = 0$

$or,\ 2m^2(m-1) + 2m(m-1) - 1(m-1) = 0$

or, $(m-1)(2m^2 + 2m - 1) = 0$

$\therefore\ m = 1, m = -1 \pm \sqrt{3}$

General solution is

$y = c_1 e^x + c_2 e^{(-1+\sqrt{3})x} + c_3 e^{(-1-\sqrt{3})x}$

or, $y = c_1 e^x + e^{-x}(c_2 e^{\sqrt{3}x} + c_3 e^{-\sqrt{3}x})$

Ex-4: Solve $\frac{d^3y}{dx^3} - 3\frac{dy}{dx} + 2y = 0$

Solu: Given equation is

$\frac{d^3y}{dx^3} - 3\frac{dy}{dx} + 2y = 0$.....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore\ \frac{dy}{dx} = me^{mx},\ \frac{d^2y}{dx^2} = m^2e^{mx},\ \frac{d^3y}{dx^3} = m^3e^{mx}$

Then equation (i) becomes

$m^3e^{mx} - 3me^{mx} + e^{mx} = 0$

or, $(2m^3 - 3m + 2)e^{mx} = 0$

Since, $e^{mx} \neq 0$

$\therefore$ The auxiliary equation is

$m^3 - 3m + 2 = 0$

$or,\quad m^3 - m^2 + m^2 - m - 2m + 2 = 0$

$or,\ m^2(m-1) + m(m-1) - 2(m-1) = 0$

$or, (m-1)(m^2 + m - 2) = 0$

$or, (m-1)(m+2)(m-1) = 0$

$\therefore\ m = 1, 1, -2$

$\therefore$ **General solution is** $y = (c_1 + c_2 x)e^x + c_3 e^{-2x}$

Ex-5: Solve $\frac{d^4y}{dx^4} + 4y = 0$

Solu: Given equation is

$\frac{d^4y}{dx^4} + 4y = 0$.....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore\ \frac{dy}{dx} = me^{mx},\ \frac{d^2y}{dx^2} = m^2e^{mx},\ \frac{d^3y}{dx^3} = m^3e^{mx},\ \frac{d^4y}{dx^4} = m^4e^{mx}$

Then equation (i) becomes

$m^4e^{mx} + 4e^{mx} = 0$ Or, $(m^4 + 4)e^{mx} = 0$ Since, $e^{mx} \neq 0$

$\therefore$ The auxiliary equation is

$m^4 + 4 = 0$

or, $(m^2 + 2)^2 - (2m)^2 = 0$

Or, $(m^2 + 2m + 2)(m^2 - 2m + 2) = 0$

$\therefore\ m^2 + 2m + 2 = 0$

$m^2 - 2m + 2 = 0$

$\therefore\ m = 1 \pm i,\ -1 \pm i$

$\therefore$ **General solution is** $y = e^x(c_1 \cos x + c_2 \sin x) + e^{-x}(c_3 \cos x + c_4 \sin x)$

Ex-: 6 Find the particular solution of $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0;\ y(0) = 0$ and $y'(0) = 1$

Solu: Given equation is $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$.....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore\ \frac{dy}{dx} = me^{mx},\ \frac{d^2y}{dx^2} = m^2e^{mx}$

Then equation (i) becomes

$m^2e^{mx} + 3me^{mx} + 2e^{mx} = 0$

or, $(m^2 + 3m + 2)e^{mx} = 0$

Since, $e^{mx} \neq 0$

The auxiliary equation is

$m^2 + 3m + 2 = 0 \quad \therefore\ m = -1, -2$

General solution is

$y = c_1 e^{-x} + c_2 e^{-2x}$

$\therefore y' = -c_1 e^{-x} - 2c_2 e^{-2x}$

Now $y(0) = 0$

$0 = c_1 + c_2 \quad$ or, $c_1 = -c_2$

$y'(0) = 1$

$1 = -c_1 - 2c_2$

Solving we get,

$c_1 = 1,\ c_2 = -1$

$\therefore$ Particular solution is $y = e^{-x} - e^{-2x}$

Ex-: 7 Find the particular solution of $(D^2 + 4D + 8)y = 0$; $y(0) = 0$, $y'(0) = 8$

Solu: Given equation is

$(D^2 + 4D + 8)y = 0$.....(i)

Let $y = e^{mx}$ be a solution of equation (i)

$\therefore Dy = me^{mx}, \quad D^2 y = m^2 e^{mx}$

Then equation (i) becomes

$m^2 e^{mx} + 4me^{mx} + 8e^{mx} = 0$

or, $(m^2 + 4m + 8)e^{mx} = 0$

Since, $e^{mx} \neq 0$

$\therefore$ The auxiliary equation is

$m^2 + 4m + 8 = 0$

$\therefore\ m = -2 \pm 2i$

$\therefore$ General solution is

$y = e^{-2x}(c_1 \cos 2x + c_2 \sin 2x) \qquad \therefore$

$y' = -2e^{-2x}(c_1 \cos 2x + c_2 \sin 2x) + e^{-2x}(-2c_1 \sin 2x + 2c_2 \cos 2x)$

Now $y(0) = 0$

$\therefore \; c_1 = 0$

$y'(0) = 8,$

$\therefore \; 8 = -2c_1 + 2c_2$

$or, \quad c_2 = 4$

$\therefore$ Particular solution is $y = 4e^{-2x}\sin 2x$

Home Work:

HW-1: Solve $(D^2 + 5D + 4)y = 0$ Ans: $y = c_1e^{-x} + c_2e^{-4x}$

HW-2: Solve $(D^2 + 6D + 9)y = 0$ Ans: $y = (c_1 + c_2x)e^{-3x}$

HW-3: Solve $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$ Hnits: $m = -2 \pm 3i$

Ans: $y = e^{-2x}(c_1 \cos 3x \pm c_2 \sin 3x)$